

Regularity of Minimal Surface Equations

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Some notations:

Ω is always a bounded open domain in $\mathbb{R}^n$.

$C^k(\Omega)$ is the set of functions with continuous derivatives up to order k in Ω , where $k \in \mathbb{Z}_+ \cup \{0\}$.

$C^{k,\alpha}(\Omega)$ is the set of functions in $C^k(\Omega)$ whose k -th derivatives are α -continuous in Ω , where $k \in \mathbb{Z}_+ \cup \{0\}$, $\alpha \in (0, 1)$.

$C^k(\bar{\Omega})$ is the set of functions with bounded continuous derivatives up to order k in Ω , where $k \in \mathbb{Z}_+ \cup \{0\}$.

$C^{k,\alpha}(\bar{\Omega})$ is the set of functions in $C^k(\bar{\Omega})$ whose k -th derivatives are uniformly α -continuous in Ω , where $k \in \mathbb{Z}_+ \cup \{0\}$, $\alpha \in (0, 1)$.

1 Regularity of Minimal Surface Equations

1.1 Introduction

We present the following two regularity theorems of minimal surface equations - the first one is about the interior regularity, while the second one is about the global regularity:

Theorem 1.1 (Interior Regularity). *Let $B = B_1$ be the unit open ball in $\mathbb{R}^n$, and suppose that $u \in W^{1,2}(B) \cap W^{1,\infty}(B)$ is a weak solution of the minimal surface equations in B in the following sense:*

$$\int_B \frac{1}{\sqrt{1 + |\nabla u|^2}} \nabla u \cdot \nabla \phi = \int_B f \phi, \quad \forall \phi \in W_0^{1,2}(B),$$

and suppose that $f \in C^\infty(\bar{B})$, then $u \in C^\infty(B)$.

Theorem 1.2 (Global Regularity). *Let $\Omega \subset \mathbb{R}^n$ be a bounded open domain with C^∞ boundary. Suppose that $u \in W^{1,2}(\Omega) \cap W^{1,\infty}(\Omega)$ is a weak solution of the minimal surface equations in B , and there exists a smooth function $\phi \in C^\infty(\bar{\Omega})$ such that $u - \phi \in W_0^{1,2}(\Omega)$, then $u \in C^\infty(\bar{\Omega})$.*

1.2 Interior Regularity

In this section, we will prove Theorem 1.1. The proof contains four steps:

- (1) From $W^{1,\infty}$ to $W^{2,2}$;
- (2) From $W^{2,2}$ to $C^{1,\alpha}$;
- (3) From $C^{1,\alpha}$ to $C^{2,\alpha}$;
- (4) From $C^{2,\alpha}$ to C^∞ .

All the results are interior.

1.2.1 Step 1: From $W^{1,\infty}$ to $W^{2,2}$

One might expect to use the perturbation techniques in the linear theory to obtain $C^{1,\alpha}$ -regularity of the solution, given the Hölder-continuous coefficients. However, since the term $\frac{1}{\sqrt{1+|\nabla u|^2}}$ is only a bounded measurable function, we cannot directly use the linear theory. Instead, we will first prove the $W^{2,2}$ -regularity of the solution.

Proposition 1.3. *In the assumption of Theorem 1.1, for any $0 < r < 1$, $u \in W^{2,2}(B_r)$.*

Proof. For all $0 < r < 1$, we choose $R \in (r, 1)$.

For any $u \in L^1(B_R)$, we define the difference quotient of u by:

$$\delta_i^h u(x) = \frac{u(x + he_i) - u(x)}{h}, \quad \forall x \in B_R, \quad 0 < |2h| < 1 - R.$$

Now for all $1 \leq k \leq n$, let the test function $\phi = \delta_k^{-h}(\eta^2 \delta_k^h u)$, where $\eta \in C_0^\infty(B_R)$. Let $F^i = \frac{p_i}{\sqrt{1+|p|^2}}$, $p \in \mathbb{R}^n$, then we have

$$\int_{B_R} F^i(\nabla u) \partial_i (\delta_k^{-h}(\eta^2 \delta_k^h u)) = - \int_{B_R} \delta_k^h F^i(\nabla u) \partial_i (\eta^2 \delta_k^h u).$$

Using the weak solution equation, we have

$$- \int_{B_R} \delta_k^h F^i(\nabla u) \partial_i (\eta^2 \delta_k^h u) = \int_{B_R} f \delta_k^{-h}(\eta^2 \delta_k^h u).$$

To express $\delta_k^h F^i$, we have

$$\begin{aligned} \delta_k^h F^i(x) &= \frac{F^i(\nabla u(x + he_k)) - F^i(\nabla u(x))}{h} \\ &= \frac{1}{h} \int_0^1 \frac{d}{dt} F^i(\nabla u(x) + th \delta_k^h \nabla u(x)) dt \\ &= \partial_j \delta_k^h u \int_0^1 \partial_{p_j} F^i(\nabla u(x) + th \delta_k^h \nabla u(x)) dt \\ &= \theta_{ij} \partial_j \delta_k^h u, \end{aligned}$$

where $\theta_{ij} = \int_0^1 \partial_{p_j} F^i(\nabla u(x) + t h \delta_k^h \nabla u(x)) dt$. By the assumption we can deduce that there exist $c, C > 0$ such that for all $\xi \in \mathbb{R}^n$, we have

$$c |\xi|^2 \leq \theta_{ij} \xi_i \xi_j \leq C |\xi|^2.$$

Then we have

$$- \int_{B_R} \theta_{ij} \partial_j \delta_k^h u \partial_i (\eta^2 \delta_k^h u) = \int_{B_R} f \delta_k^{-h} (\eta^2 \delta_k^h u).$$

Let $v = \delta_k^h u$, then

$$\int_{B_R} \theta_{ij} \partial_j v \partial_i (\eta^2 v) = - \int_{B_R} f \delta_k^{-h} (\eta^2 v),$$

We choose $r_1, r_2 > 0$ such that $r < r_1 < r_2 < R$, and we always assume that $|2h| < \min\{1 - R, r_1 - r, r_2 - r_1, R - r_2\}$. Now if $\eta \in C_0^\infty(B_{r_1})$, then

$$\begin{aligned} \int_{B_r} |\nabla v|^2 &\leq \int_{B_{r_1}} |\nabla (\eta v)|^2 \\ &\leq \frac{1}{c} \int_{B_{r_1}} \theta_{ij} \partial_i (v \eta) \partial_j (v \eta) \\ &= \frac{1}{c} \int_{B_{r_1}} \theta_{ij} (v^2 \partial_i \eta \partial_j \eta + 2 \eta v \partial_j v \partial_i \eta + \eta^2 \partial_i v \partial_j v). \end{aligned}$$

Using the fact that

$$\int_{B_{r_1}} \theta_{ij} \partial_j v (2 \eta v \partial_i \eta + \eta^2 \partial_i v) = - \int_{B_{r_2}} f \delta_k^{-h} (\eta^2 v) = \int_{B_{r_2}} \delta_k^h f \eta^2 v,$$

we have

$$\begin{aligned} \int_{B_{r_1}} |\nabla v|^2 &\leq \frac{1}{c} \int_{B_{r_1}} \theta_{ij} v^2 \partial_i \eta \partial_j \eta + \frac{1}{c} \int_{B_{r_2}} \delta_k^h f \eta^2 v \\ &\leq \frac{C}{c} \int_{B_{r_1}} v^2 |\nabla \eta|^2 + \frac{|\nabla f|_{L^\infty}}{c} \int_{B_{r_2}} |\eta^2 v| \\ &\leq \frac{C |\nabla \eta|_{L^\infty}^2}{c} \int_{B_R} v^2 + \frac{|\nabla f|_{L^\infty}}{c} \int_{B_R} |v| \\ &\leq \tilde{C} \left(\|u\|_{W^{1,2}(B_R)}^2 + \|u\|_{W^{1,1}(B_R)} \right) \end{aligned}$$

Note that RHS is independent of h , we have $\partial_k u \in W^{1,2}(B_{r_1})$. Therefore $u \in W^{2,2}(B_R)$. $\square$

Lemma 1.4 (Property of F). *For the function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $F^i(p) = \frac{p_i}{\sqrt{1+|p|^2}}$, for all $K > 0$, in the domain $D = \{p \in \mathbb{R}^n \mid |p| < K\}$, there exist positive constants $\lambda, \Lambda > 0$ such that for all $\xi \in \mathbb{R}^n$,*

$$\partial_j F^i = \partial_i F^j, \tag{1}$$

$$\lambda |\xi|^2 \leq \partial_j F^i(p) \xi_i \xi_j \leq \Lambda |\xi|^2. \tag{2}$$

Proof.

$$\partial_j F^i = \frac{\delta_{ij} \sqrt{1 + |p|^2} - p_i p_j / \sqrt{1 + |p|^2}}{1 + |p|^2} = \partial_i F^j,$$

so (1) is proved. For (2), note that for all $\xi \in \mathbb{R}^n$,

$$\partial_j F^i \xi_i \xi_j = \frac{1}{\sqrt{1 + |p|^2}} |\xi|^2 - \frac{1}{(1 + |p|^2)^{\frac{3}{2}}} \left(\sum_{i=1}^n p_i \xi_i \right)^2,$$

on the one hand,

$$\partial_j F^i \xi_i \xi_j \leq \frac{1}{\sqrt{1 + |p|^2}} |\xi|^2 \leq |\xi|^2.$$

On the other hand,

$$\begin{aligned} \partial_j F^i \xi_i \xi_j &\geq \frac{1}{\sqrt{1 + |p|^2}} |\xi|^2 - \frac{1}{(1 + |p|^2)^{\frac{3}{2}}} |p|^2 |\xi|^2 \\ &= \frac{1}{(1 + |p|^2)^{\frac{3}{2}}} |\xi|^2 \\ &\geq \frac{1}{(1 + K^2)^{\frac{3}{2}}} |\xi|^2. \end{aligned}$$

□

Other results about the difference quotient are presented below:

Lemma 1.5 (Gilbarg & Trudinger, Lemma 7.23). *For any open bounded domain $\Omega \subset \mathbb{R}^n$, if $u \in W^{1,p}(\Omega)$, $1 \leq p < \infty$, then for any subdomain $\Omega' \Subset \Omega$, if $|h| < \text{dist}(\Omega', \partial\Omega)$, then*

$$\|\delta_i^h u\|_{L^p(\Omega')} \leq \|\partial_i u\|_{L^p(\Omega)}.$$

Lemma 1.6 (Gilbarg & Trudinger, Lemma 7.24). *For any open bounded domain $\Omega \subset \mathbb{R}^n$, let $u \in L^p(\Omega)$, $1 < p < \infty$. Suppose that for any subdomain $\Omega' \Subset \Omega$, there exists a constant $K > 0$ such that for all $1 \leq i \leq n$, $|h| \leq \text{dist}(\Omega', \partial\Omega)$,*

$$\|\delta_i^h u\|_{L^p(\Omega')} \leq K,$$

then all of the weak derivative $\partial_i u$ exist, and they satisfy

$$\|\partial_i u\|_{L^p(\Omega)} \leq K.$$

1.2.2 Step 2: From $W^{2,2}$ to $C^{1,\alpha}$

Now we use the non-perturbation method (not relies on the continuity of the coefficients) to obtain the $C^{0,\alpha}$ -regularity of the weak derivative of the solution u .

Proposition 1.7. *In the assumption of Theorem 1.1, for any $0 < r < 1$, $u \in C^{1,\alpha}(B_r)$.*

For any $0 < r < 1$, we choose $R \in (r, 1)$. Now for any $1 \leq k \leq n$, we take the test function to be $\partial_k \phi$, $\phi \in W_0^{2,2}(B_R)$ in the weak solution equation, we have

$$\int_{B_R} F^i \partial_{ik} \phi = \int_{B_R} f \partial_k \phi.$$

Integrating by parts, we obtain

$$\int_{B_R} \partial_j F^i \partial_j (\partial_k u) \cdot \partial_i \phi = \int_{B_R} \partial_k f \cdot \phi,$$

which means if we take $v = \partial_k u$, $\tilde{f} = \partial_k f$, $\tilde{a}_{ij} = \partial_j F^i$, then v satisfies following equation:

$$\int_{B_R} \tilde{a}_{ij} \partial_j v \cdot \partial_i \phi = \int_{B_R} \tilde{f} \phi, \quad \forall \phi \in C_0^\infty(B_R).$$

By approximation, the equation above holds for any test function $\phi \in W_0^{1,2}(B_R)$.

Because $u \in W^{1,\infty}(B_1)$, we have $v \in L^\infty(B_1)$.

First, we have to obtain the local boundedness of the weak subsolution,

Theorem 1.8 (Local boundedness). *Assume $a_{ij} \in L^\infty(B_1)$ satisfying*

$$a_{ij}(x) \xi_i \xi_j \geq \lambda |\xi|^2, \quad \forall x \in B_1, \xi \in \mathbb{R}^n,$$

$$|a_{ij}|_{L^\infty} \leq \Lambda,$$

for some $\lambda, \Lambda > 0$. And suppose that $u \in W^{1,2}(B_1)$ is the subsolution in the following sense

$$\int_{B_1} a_{ij} \partial_i u \partial_j \phi \leq \int_{B_1} f \phi, \quad \forall \phi \in H_0^1(B_1), \phi \geq 0 \text{ in } B_1,$$

Then if $f \in L^\infty(B_1)$ we have $u^+ \in L_{loc}^\infty(B_1)$. Moreover, for arbitrary $\theta \in (0, 1)$ and $p > 0$,

$$\sup_{B_\theta} u^+ \leq C \left\{ \frac{1}{(1-\theta)^{\frac{n}{2}}} \|u^+\|_{L^2(B_1)} + \|f\|_{L^\infty(B_1)} \right\},$$

where $C = C(n, \lambda, \Lambda, q)$ is a positive constant.

The proof is based on De-Giorgi iteration.

Proof. We prove the theorem in the case that $\theta = \frac{1}{2}$.

For $k \geq 0$, we define $v = (u - k)^+$. For arbitrary $\zeta \in C_0^1(B_1)$, suppose the test function to be $\phi = v\zeta^2$. Then

$$\begin{aligned}
& \int_{u>k} a_{ij} \partial_i u \partial_j \phi \\
&= \int_{u>k} (a_{ij} \zeta^2 \partial_i u \partial_j v + 2a_{ij} v \zeta \partial_i u \partial_j \zeta) \\
&\geq \left(\int_{u>k} \lambda |\nabla v|^2 \zeta^2 - 2\Lambda |\nabla v| |\nabla \zeta| \right) \\
&\geq \lambda^2 \int_{u>k} \zeta^2 |\nabla v|^2 - 2\Lambda \left(\int_{u>k} \frac{\lambda^2}{4\Lambda} |\nabla v|^2 \zeta^2 + \frac{\Lambda}{\lambda^2} \int_{u>k} |\nabla \zeta|^2 v^2 \right) \\
&= \frac{\lambda^2}{2} \int_{u>k} \zeta^2 |\nabla v|^2 - \frac{2\Lambda^2}{\lambda^2} \int_{u>k} |\nabla \zeta|^2 v^2.
\end{aligned}$$

Therefore,

$$\begin{aligned}
& \int_{u>k} |\nabla v|^2 \zeta^2 \\
&\leq C \int_{u>k} (|\nabla \zeta|^2 v^2 + |f| v \zeta^2) \\
&= C \int_{u>k} (|\nabla \zeta|^2 v^2 + |f| v \zeta^2) \\
&\leq C \int_{u>k} (|\nabla \zeta|^2 v^2 + |f| v \zeta^2).
\end{aligned}$$

Then we have

$$\int_{u>k} |\nabla(v\zeta)|^2 \leq C \int_{u>k} (|\nabla \zeta|^2 v^2 + |f| v \zeta^2).$$

By Sobolev inequality for $v\zeta \in W_0^{1,2}(B_1)$, we have

$$\left(\int_{u>k} (v\zeta)^{2^*} \right)^{\frac{2}{2^*}} \leq c(n) \int_{B_1} |\nabla(v\zeta)|^2,$$

where

$$2^* = \begin{cases} \frac{2n}{n-2}, & n \geq 3, \\ \text{any real number} > 2, & n = 2. \end{cases}$$

Then for small $\delta > 0$ and $\zeta \leq 1$, we have

$$\begin{aligned}
\int_{u>k} |f| v \zeta^2 &\leq \|f\|_{L^\infty} \left(\int_{u>k} |v \zeta|^{2^*} \right)^{\frac{1}{2^*}} |\{v \zeta \neq 0\}|^{1-\frac{1}{2^*}} \\
&\leq C(n) \|f\|_{L^\infty} \left(\int_{u>k} |\nabla(v \zeta)|^2 \right)^{\frac{1}{2}} |\{v \zeta \neq 0\}|^{1-\frac{1}{2^*}} \\
&\leq \delta \int_{u>k} |\nabla(v \zeta)|^2 + C(n, \delta) \|f\|_{L^\infty}^2 |\{v \zeta \neq 0\}|^{2-\frac{2}{2^*}} \\
&\leq \delta \int_{u>k} |\nabla(v \zeta)|^2 + C(n, \delta) \|f\|_{L^\infty}^2 |\{v \zeta \neq 0\}|.
\end{aligned}$$

Note that we use $|\{v \zeta \neq 0\}| \leq \omega_n$ and $2 - \frac{2}{2^*} > 1$ (equivalent to $2^* > 2$) above.

Therefore, we have following estimates:

$$\int_{u>k} |\nabla(v \zeta)|^2 \leq C \int_{u>k} v^2 |\nabla \zeta|^2 + CF^2 |\{v \zeta \neq 0\}|,$$

where $F = \|f\|_{L^\infty(B_1)}$.

Now we have

$$\begin{aligned}
\int_{u>k} v^2 \zeta^2 &\leq \left(\int_{u>k} (v \zeta)^{2^*} \right)^{\frac{2}{2^*}} |\{v \zeta \neq 0\}|^{1-\frac{2}{2^*}} \\
&\leq C(n) \int_{u>k} |\nabla(v \zeta)|^2 |\{v \zeta \neq 0\}|^{1-\frac{2}{2^*}} \\
&\leq C \left(\int_{u>k} v^2 |\nabla \zeta|^2 |\{v \zeta \neq 0\}|^{1-\frac{2}{2^*}} + F^2 |\{v \zeta \neq 0\}|^{2-\frac{2}{2^*}} \right) \\
&\leq C \left(\int_{u>k} v^2 |\nabla \zeta|^2 |\{v \zeta \neq 0\}|^\epsilon + F^2 |\{v \zeta \neq 0\}|^{1+\epsilon} \right),
\end{aligned}$$

where we choose a small constant ϵ . We choose a cut-off function: for all $0 < r < R \leq 1$, take $\zeta \in C_0^\infty(B_R)$ such that $\zeta = 0$ in B_1 , $0 \leq \zeta \leq 1$, and $|\nabla \zeta| \leq \frac{2}{R-r}$ holds in B_1 .

Let $A(k, r) = \{x \in B_r : u \geq k\}$, then for any $0 < r < R \leq 1$, $k > 0$, we have

$$\int_{A(k, r)} (u - k)^2 \leq C \left(\frac{1}{(R - r)^2} |\{A(k, R)\}|^\epsilon \int_{A(k, R)} (u - k)^2 + F^2 |A(k, R)|^{1+\epsilon} \right). \quad (**)$$

Now we are going to prove that there exists k such that

$$\int_{A(k, \frac{1}{2})} (u - k)^2 = 0,$$

so that $u \leq k$ which is what we want.

Take arbitrary $h > k$, $0 < r < 1$. Obviously $A(k, r) \supset A(h, r)$. So

$$\int_{A(h, r)} (u - h)^2 \leq \int_{A(k, r)} (u - k)^2,$$

and

$$|A(h, r)| = |B_r \cap \{u - k > h - k\}| \leq \frac{1}{(h - k)^2} \int_{A(k, r)} (u - k)^2.$$

Therefore, by (**), for arbitrary $h > k \geq k_0$, $\frac{1}{2} \leq r < R \leq 1$, we have

$$\begin{aligned} & \int_{A(h, r)} (u - k)^2 \\ & \leq C \left(\frac{1}{(R - r)^2} \int_{A(h, R)} (u - h)^2 + F^2 |A(h, R)| \right) |A(h, R)|^\epsilon \\ & \leq C \left(\frac{1}{(R - r)^2} + \frac{F^2}{(h - k)^2} \right) \frac{1}{(h - k)^{2\epsilon}} \left(\int_{A(k, R)} (u - k)^2 \right)^{1+\epsilon}. \end{aligned}$$

i.e.

$$\|(u - h)^+\|_{L^2(B_r)} \leq C \left(\frac{1}{R - r} + \frac{F}{h - k} \right) \frac{1}{(h - k)^\epsilon} \|(u - k)^+\|_{L^2(B_R)}^{1+\epsilon}.$$

Iteration: let $\phi(k, r) = \|(u - k)^+\|_{L^2(B_r)}$.

Take $k_0 = \|u^+\|_{L^2}$, and suppose that $k > k_0$ whose exact value will be determined below. For each $l = 0, 1, \dots$, we define

$$\begin{aligned} k_l &= k_0 + k \left(1 - \frac{1}{2^l} \right) (\leq k_0 + k), \\ r_l &= \tau + \frac{1}{2^l} (1 - \tau). \end{aligned}$$

Then

$$\begin{aligned} k_l - k_{l-1} &= \frac{k}{2^l}, \\ r_{l-1} - r_l &= \frac{1}{2^l} (1 - \tau). \end{aligned}$$

So for any non-negative integer l ,

$$\begin{aligned} \phi(k_l, r_l) &\leq C \left(\frac{2^l}{1 - \tau} + \frac{2^l F}{k} \right) \frac{2^{\epsilon l}}{k^\epsilon} \phi(k_{l-1}, r_{l-1})^{1+\epsilon} \\ &\leq \frac{C}{1 - \tau} \cdot \frac{F}{k^{1+\epsilon}} \cdot 2^{(1+\epsilon)l} \phi(k_{l-1}, r_{l-1})^{1+\epsilon}. \end{aligned}$$

We now prove by induction that for any non-negative integer l , we have the following exponential estimate

$$\phi(k_l, r_l) \leq \frac{\phi(k_0, r_0)}{\gamma^l} \quad (***)$$

holds for some $\gamma > 1$ if k is arbitrarily big. When $l = 0$, it is trivial. Assume the conclusion holds for $l - 1$,

$$\phi(k_{l-1}, r_{l-1})^{1+\epsilon} \leq \left(\frac{\phi(k_0, r_0)}{\gamma^{l-1}} \right)^{1+\epsilon} = \frac{\phi(k_0, r_0)^\epsilon}{\gamma^{l\epsilon-(1+\epsilon)}} \cdot \frac{\phi(k_0, r_0)}{\gamma^l}.$$

Then

$$\phi(k_l, r_l) \leq \frac{C\gamma^{1+\epsilon}}{1-\tau} \cdot \frac{F}{k^{1+\epsilon}} \cdot \phi(k_0, r_0)^\epsilon \cdot \frac{2^{l(1+\epsilon)}}{\gamma^{l\epsilon}} \cdot \frac{\phi(k_0, r_0)}{\gamma^l}.$$

We choose γ such that $\gamma^\epsilon = 2^{1+\epsilon}$. Also, we need to check

$$\frac{C\gamma^{1+\epsilon}}{1-\tau} \cdot \left(\frac{\phi(k_0, r_0)}{k} \right)^\epsilon \cdot \frac{F}{k} \leq 1.$$

It is enough to choose $k = T(F + \phi(k_0, r_0))$, where $T > 0$ is big.

In (***), let $l \rightarrow \infty$, we obtain

$$\phi(k_0 + k, \tau) = 0,$$

i.e.

$$\sup_{B_{\frac{1}{2}}} u^+ \leq (T+1)(k_0 + F + \phi(k_0, r_0)) \leq C(T+1)(\|u^+\|_{L^2(B_1)} + F).$$

□

Now we can concentrate on the Hölder-regularity of v . First we shall prove this result in the homogeneous case where $\tilde{f} = 0$.

Theorem 1.9. *If $s \in W^{1,2}(B)$ satisfying*

$$\int_B \tilde{a}_{ij} \partial_j s \partial_i \phi = 0, \forall \phi \in W_0^{1,2}(B),$$

then $s \in C^\alpha(B)$.

Proof. For any $0 < R < 1$, by the local boundedness of the solution, let

$$\alpha_1 = \sup_{B_R} s, \quad \beta_1 = \inf_{B_R} s.$$

Suppose that

$$A = \{x \in B_R : s(x) \geq \frac{\alpha_1 + \beta_1}{2}\}, \quad B = \{x \in B_R : s(x) \leq \frac{\alpha_1 + \beta_1}{2}\},$$

then $|A| \geq \frac{1}{2}|B_R|$ or $|B| \geq \frac{1}{2}|B_R|$. Without generality, we can assume that $|A| \geq \frac{1}{2}|B_R|$.

Let $r(x) = 2 \cdot \frac{s(x) - \beta_1}{\alpha_1 - \beta_1}$ be a non-negative function in B_R . For any $\delta > 0$, let $\tilde{r} = r + \delta \geq \delta$, then we assert that for $l = (\log \tilde{r})^-$, for any non-negative test function $\phi \in W_0^{1,2}(B_R)$,

$$\int_{B_R} \tilde{a}^{ij} \partial_j l \partial_i \phi \leq 0.$$

By Theorem 1.8,

$$\sup_{B_{\frac{R}{2}}} l \leq C \left(\int_{B_R} |l|^2 \right)^{\frac{1}{2}}. \quad (3)$$

Let $A_\delta = \{x \in B_R : l = 0\} = \{x \in B_R : r \geq 1 - \delta\}$, so

$$|A_\delta| \geq |\{x \in B_R : r \geq 1\}| = |A| \geq \frac{1}{2} |B_R|$$

If there exists a sequence $\{u_m\} \subset W^{1,2}(B_R)$ such that

$$|\{x \in B_R : u_m = 0\}| \geq \frac{1}{2} |B_R|, \quad \int_{B_R} u_m^2 = 1, \quad \int_{B_R} |\nabla u_m|^2 \rightarrow 0 (m \rightarrow \infty),$$

we can take a subsequence (we still name it $\{u_m\}$) such that $u_m \rightarrow u_0 \in W^{1,2}(B_R)$ strongly in $L^2(B_R)$ and weakly in $W^{1,2}(B_R)$. It is trivial that u_0 is a non-zero constant K , so

$$0 = \lim_{m \rightarrow \infty} \int_{B_R} |u_m - u_0|^2 \geq \lim_{m \rightarrow \infty} \int_{u_m=0} |u_m - u_0|^2 \geq \frac{K^2}{2} |B_R|,$$

a contradiction. So there exists a constant $C > 0$ such that for all $u \in W^{1,2}(B_R)$ satisfying

$$|\{x \in B_R : u(x) = 0\}| \geq \frac{1}{2} |B_R|,$$

we have

$$\int_{B_R} u^2 \leq C \int_{B_R} |\nabla u|^2.$$

Especially,

$$\int_{B_R} |l|^2 \leq C \int_{B_R} |\nabla l|^2. \quad (4)$$

Next we aim to prove that $\int_{B_R} |\nabla l|^2$ is bounded. Take test function $\phi = \frac{\zeta^2}{\tilde{r}}, C_0^1(B_R)$,

$$0 \leq \int_{B_R} \tilde{a}_{ij} \partial_j \tilde{r} \partial_i \left(\frac{\zeta^2}{\tilde{r}} \right) = - \int_{B_R} \zeta^2 \frac{\tilde{a}_{ij} \partial_i \tilde{r} \partial_j \tilde{r}}{\tilde{r}^2} + 2 \int_{B_R} \frac{\zeta \tilde{a}_{ij} \partial_i \tilde{r} \partial_j \zeta}{\tilde{r}},$$

which is equivalent to

$$\int_{B_R} \zeta^2 \tilde{a}_{ij} \partial_i l \partial_j l \leq -2 \int_{B_R} \zeta \tilde{a}_{ij} \partial_i l \partial_j \zeta.$$

Therefore,

$$\begin{aligned} \lambda \int_{B_R} \zeta^2 |\nabla l|^2 &\leq 2\Lambda \int_{B_R} |\zeta| |\nabla l| |\nabla \zeta| \\ &\leq \frac{\lambda}{2} \int_{B_R} \zeta^2 |\nabla l|^2 + \frac{2\Lambda^2}{\lambda} \int_{B_R} |\nabla \zeta|^2, \end{aligned}$$

which gives

$$\int_{B_R} \zeta^2 |\nabla l|^2 \leq C \int_{B_R} |\nabla \zeta|^2.$$

Fix $\zeta \in C_0^1(B_r)$ such that $\zeta = 1$ in $B_{\frac{R}{2}}$, so we obtain

$$\int_{B_{\frac{R}{2}}} |\nabla l|^2 \leq C.$$

Combining it with (3)(4), we finally obtain $\inf_{B_{\frac{R}{2}}} (r + \delta) \geq e^{-C}$. Note that C is independent of δ , let $\delta \rightarrow 0$,

$$\inf_{B_{\frac{R}{2}}} r \geq \tilde{C} > 0,$$

i.e.

$$\inf_{B_{\frac{R}{2}}} s \geq \frac{\tilde{C}}{2} \operatorname{osc}_{B_R} s + \inf_{B_R} s.$$

Let $\alpha_2 = \sup_{B_{\frac{R}{2}}} s$, $\beta_2 = \inf_{B_{\frac{R}{2}}} s$, $\kappa = \min\{\frac{1}{2}, 2\tilde{C}\} < 1$, then

$$\beta_2 \geq \kappa(\alpha_1 - \beta_1) + \beta_1,$$

so that

$$\alpha_2 - \beta_2 \leq \alpha_1 - \beta_2 \leq (1 - \kappa)(\alpha_1 - \beta_1).$$

Using scaling technique, for all $0 < r < 1$, we can deduce that there exists a constant γ independent of R such that for all $x_0 \in B_{\frac{r+1}{2}}$, $0 < 2R < \operatorname{dist}(x_0, \partial B_{\frac{r+1}{2}})$,

$$\operatorname{osc}_{B_{\frac{R}{2}}(x_0)} s \leq \gamma \operatorname{osc}_{B_R(x_0)} s.$$

Now for all $x, y \in B_r$, if $|x - y| \geq \frac{1-R}{10000}$, we have

$$|s(x) - s(y)| \leq \frac{20000 |s|_{L^\infty}}{1-R} |x - y| \leq \frac{20000 |s|_{L^\infty}}{1-R} |x - y|^\alpha, \quad \forall \alpha \in (0, 1);$$

If $|x - y| < \frac{1-R}{10000}$, then we choose $v = 2|x - y|$ and let N be a positive number such that $2^N v \leq \frac{1-R}{2} < 2^{N+1} v$, then $B_{2^N v}(x) \subset B_{\frac{1+R}{2}}$. From $N > \log_2(\frac{1-R}{v}) - 2$, we have

$$\begin{aligned} |s(x) - s(y)| &\leq \operatorname{osc}_{B_v(x)} s \leq \gamma^N \operatorname{osc}_{B_{2^N v}(x)} s \leq \gamma^N \operatorname{osc}_{B_{\frac{1+R}{2}}} s \\ &\leq 2\gamma^{\log_2((1-R)/v)-2} \sup_{B_{\frac{1+R}{2}}} s = \frac{1}{\gamma^2} (1-R)^{\log_2 \gamma} \sup_{B_{\frac{1+R}{2}}} s \cdot |x - y|^{-\log_2 \gamma}. \end{aligned}$$

Therefore, take $\alpha = \min\{\frac{1}{2}, -\log_2 \gamma\}$, we obtain $u \in C^\alpha(B_r)$. Moreover, $s \in C^\alpha(B)$ since r is arbitrary. $\square$

Remark 1.10. From the proof of Theorem 1.9, combining with the results of Theorem 1.8, we have

$$|s|_{C^\alpha(B)} \leq C(n, \lambda, \Lambda) \|s\|_{L^2(B)}.$$

Now we are going to obtain the inhomogeneous results:

Theorem 1.11. If $v \in W^{1,2}(B)$ satisfying

$$\int_B \tilde{a}_{ij} \partial_j v \partial_i \phi = \int_B \tilde{f} \phi, \quad \forall \phi \in W_0^{1,2}(B),$$

where $\tilde{f} \in C^\infty(B)$, then $v \in C^\alpha(B)$ for some $\alpha \in (0, \frac{1}{2})$.

Proof. Given any $0 < r < 1$, for all $x \in B_{\frac{1+R}{2}}$, take $R_0 = \frac{1-r}{8}$. Then for any $0 < R < R_0$, take $s \in W^{1,2}(B_R(x))$ such that $s - v \in W_0^{1,2}(B_R(x))$

$$\int_{B_R(x)} \tilde{a}_{ij} \partial_j s \partial_i \phi = 0, \quad \forall \phi \in W_0^{1,2}(B).$$

Let $z = v - s \in W_0^{1,2}(B_R(x))$.

1. Estimate for ∇z .

For all $0 < \rho < R$, take test function $\phi = z$,

$$\int_{B_R(x)} \tilde{a}_{ij} \partial_j (v + z) \partial_i z = \int_{B_R(x)} f z,$$

which is equivalent to

$$\int_{B_R(x)} \tilde{a}_{ij} \partial_i z \partial_j z = \int_{B_R(x)} f z.$$

Therefore,

$$\begin{aligned}
\int_{B_R(x)} |\nabla z|^2 &\leq \frac{1}{\lambda} \int_{B_R(x)} \tilde{a}_{ij} \partial_i z \partial_j z \leq \frac{1}{\lambda} \int_{B_R(x)} f z \\
&\leq \frac{1}{\lambda} \|\tilde{f}\|_{L^\infty} \left(\int_{B_R(x)} z^{2^*} \right)^{\frac{1}{2^*}} |B_R(x)|^{1-\frac{1}{2^*}} \\
&\leq \frac{C(n)}{\lambda} \cdot R^{n(1-\frac{1}{2^*})} \|\tilde{f}\|_{L^\infty} \left(\int_{B_R(x)} |\nabla z|^2 \right)^{\frac{1}{2}}
\end{aligned}$$

which gives

$$\int_{B_R(x)} |\nabla z|^2 \leq C R^{2n(1-\frac{1}{2^*})} \|\tilde{f}\|_{L^\infty}^2.$$

2. Estimate for ∇s For all $0 < R' < R_0$, Let $\tilde{s} = s - \frac{1}{|B_{R'}(x_0)|} \int_{B_{R'}(x)} s$, then by Poincare's inequality,

$$\int_{B_{R'}} |\tilde{s}|^2 \leq C(n) \int_{B_{R'}} |\nabla \tilde{s}|^2.$$

By Theorem 1.9, there exists $0 < \alpha < 1$ such that for all $y \in B_{\frac{R'}{2}}(x)$,

$$\begin{aligned}
|\tilde{s}(x) - \tilde{s}(y)|^2 &\leq C_1 |x - y|^{2\alpha} \frac{1}{R'^{n-2+2\alpha}} \|\tilde{s}\|_{L^2(B_{R'}(x))}^2 \\
&\leq C_1 |x - y|^{2\alpha} \frac{1}{R'^{n-2+2\alpha}} \|s\|_{L^2(B_{R'}(x))}^2,
\end{aligned}$$

where the constant C_1 only depends on n, λ, Λ , and the term $R^{n-2-2\alpha}$ comes from the scaling.

Now for any $0 < R < R' < \frac{R_0}{2}$, if $2R < R'$, then we can choose a cut-off function $\zeta \in C_0^\infty(B_{2R}(x))$ such that $\zeta = 1$ in $B_R(x)$, $|\nabla \zeta| \leq \frac{2}{R}$. Then take test function $\phi(y) = \zeta^2(s(y) - s(x))$,

$$\begin{aligned}
0 &= \int_{B_{2R}(x)} \tilde{a}_{ij} \partial_j s (2\zeta(s(y) - s(x)) \partial_i \zeta + \zeta^2 \partial_i s) \\
&\geq \frac{\lambda}{2} \int_{B_{2R}(x)} \zeta^2 |\nabla s|^2 - C \sup_{y \in B_R(x)} |s(y) - s(x)| \int_{B_{2R}} |\nabla \zeta|^2,
\end{aligned}$$

which gives

$$\begin{aligned}
\int_{B_R(x)} |\nabla s|^2 &\leq C \sup_{y \in B_R(x)} |s(y) - s(x)| \int_{B_{2R}} |\nabla \zeta|^2 \\
&\leq C |x - y|^{2\alpha} \frac{1}{R'^{n-2+2\alpha}} \|s\|_{L^2(B_{R'}(x))}^2 \cdot R^{n-2} \\
&\leq C \left(\frac{R}{R'} \right)^{n-2+2\alpha} \|s\|_{L^2(B_{R'}(x))}^2,
\end{aligned}$$

where C does not depend on x, R, R' .

While if $\frac{R'}{2} \leq R < R'$, then

$$\int_{B_R(x)} |\nabla s|^2 \leq 2^n \left(\frac{R}{R'} \right)^{n-2+2\alpha} \int_{B_{R'}(x)} |\nabla s|^2$$

Therefore, for any $x \in B_{\frac{1+r}{2}}$ and $0 < R < R' \leq \frac{R_0}{2} = \frac{1-r}{16}$, there exists a constant C independent of x, R, R' such that

$$\int_{B_r(x)} |\nabla s|^2 \leq C \left(\frac{R}{R'} \right)^{n-2+2\alpha} \int_{B_{R'}(x)} |\nabla s|^2.$$

3. Estimate for ∇v Now for any $x \in B_{\frac{1+r}{2}}$, $0 < R < R' < R_0$,

$$\begin{aligned} \int_{B_R(x)} |\nabla v|^2 &\leq 2 \int_{B_R(x)} |\nabla s|^2 + 2 \int_{B_R(x)} |\nabla z|^2 \\ &\leq C \left(\frac{R}{R'} \right)^{n-2+2\alpha} \int_{B_{R'}(x)} |\nabla s|^2 + 2 \int_{B_R(x)} |\nabla z|^2 \\ &\leq C \left(\left(\frac{R}{R'} \right)^{n-2+2\alpha} \int_{B_{R'}(x)} |\nabla v|^2 + \int_{B_{R'}(x)} |\nabla z|^2 \right) \\ &\leq C \left(\left(\frac{R}{R'} \right)^{n-2+2\alpha} \int_{B_{R'}(x)} |\nabla v|^2 + R'^{2n(1-\frac{1}{2^*})} \|\tilde{f}\|_{L^\infty} \right) \\ &\leq C \left(\left(\frac{R}{R'} \right)^{n-2+2\alpha} \int_{B_{R'}(x)} |\nabla v|^2 + R'^n \|\tilde{f}\|_{L^\infty} \right) \end{aligned}$$

Now we carry out a iteration technique. Let $B(t) = \frac{1}{t^{n-2+\alpha}} \int_{B_t(x)} |\nabla v|^2$, $\forall 0 < t \leq \frac{R_0}{3}$, then we have

$$B(R) \leq C \left(\left(\frac{R}{R'} \right)^\alpha B(R') + \frac{R'^n}{R^{n-2+\alpha}} \|\tilde{f}\|_{L^\infty} \right).$$

We now want to prove that $B(t)$ is bounded in $(0, \frac{R_0}{3}]$. And since if $a \leq t \leq ka$, then $B(t) \leq \frac{1}{k^{n-2+\alpha}} B(ka)$, we only need to construct a geometric sequence $a_m = q^m \frac{R_0}{3}$, $0 < q < 1$ such that for all $m \geq 0$, $B(a_m)$ is bounded. We take $q < \min\{\frac{1}{2}, (\frac{1}{2C})^{\frac{1}{\alpha}}\}$ to obtain

$$B(a_{m+1}) \leq \frac{1}{2} B(a_m) + K,$$

which easily gives $B(a_m)$ is bounded. Then there exists a constant $M > 0$ (not depending on R, R', x) such that for any $x \in B_{\frac{1+r}{2}}$, $0 < R \leq \frac{R_0}{3}$,

$$\int_{B_R(x)} |\nabla v|^2 \leq M r^{n-2+\alpha},$$

then $v \in C^{\frac{\alpha}{2}}(B_r)$. Moreover, since α is independent of r , $v \in C^{\frac{\alpha}{2}}$. $\square$

Thus from the results above, Proposition 1.7 is finally proved. Some important lemmas are listed below:

Lemma 1.12. *If $s \in W^{1,2}(\Omega)$ is a positive weak solution in the following sense:*

$$\int_{\Omega} a_{ij} \partial_j s \partial_i \phi = 0, \quad \forall \phi \in W_0^{1,2}(\Omega), \phi \geq 0,$$

then $r = (\log s)^-$ is a subsolution in the following sense:

$$\int_{\Omega} a_{ij} \partial_j r \partial_i \phi \leq 0, \quad \forall \phi \in W_0^{1,2}(\Omega), \phi \geq 0,$$

Proof. First, we prove that $\log s$ is a supersolution. For any non-negative test function $\phi \in W_0^{1,2}(\Omega)$, we have

$$\begin{aligned} \int_{\Omega} a_{ij} \partial_j (\log s) \partial_i \phi &= \int_{\Omega} a_{ij} \partial_j s \frac{\partial_i \phi}{s} \\ &= \int_{\Omega} a_{ij} \partial_j s \partial_i \left(\frac{\phi}{s} \right) + \int_{\Omega} a_{ij} \partial_j s \partial_i s \cdot \frac{\phi}{s^2} \geq 0. \end{aligned}$$

Then we prove that $(\log s)^-$ is a subsolution. Let $g = \log s$, and take $\Phi_{\epsilon}(t) = \frac{\sqrt{t^2 + \epsilon^2} - t}{2}$ for any $\epsilon > 0$, then consider $\Phi_{\epsilon}(g)$:

$$\begin{aligned} \int_{\Omega} a_{ij} \partial_j (\Phi_{\epsilon}(g)) \partial_i \phi &= \int_{\Omega} a_{ij} \Phi'_{\epsilon}(g) \partial_j g \partial_i \phi \\ &= - \int_{\Omega} a_{ij} \partial_j g \partial_i (-\Phi'_{\epsilon}(g) \phi) - \int_{\Omega} a_{ij} \partial_j g \Phi''_{\epsilon}(g) \partial_i g \cdot \phi \leq 0 \end{aligned}$$

because $\Phi'_{\epsilon}(t) = -\frac{\sqrt{t^2 + \epsilon^2} - t}{2\sqrt{t^2 + \epsilon^2}}$ is negative and bounded, and $\Phi''_{\epsilon}(t) = \frac{\epsilon^2}{(t^2 + \epsilon^2)^{\frac{3}{2}}}$ is positive. Therefore $\Phi_{\epsilon}(g)$ is a supersolution. It is not hard to prove that $\partial_i \Phi_{\epsilon}(g) \rightarrow \partial_i g^-$ almost everywhere when $g \neq 0$, and $\partial_i \Phi_{\epsilon}(g) \rightarrow -\frac{1}{2} \partial_i g$ almost everywhere when $g = 0$. Therefore by Dominated Convergence Theorem,

$$\int_{g \neq 0} a_{ij} \partial_j g^- \partial_i \phi - \frac{1}{2} \int_{g=0} a_{ij} \partial_j g \partial_i \phi \leq 0.$$

Because $\nabla g = 0$ almost everywhere in the set $\{g = 0\}$, we obtain

$$\int_{\Omega} a_{ij} \partial_j g^- \partial_i \phi \leq 0$$

for all positive test function $\phi \in W_0^{1,2}(\Omega)$. □

1.2.3 Step 3: From $C^{1,\alpha}$ to $C^{2,\alpha}$

Since the solution $u \in C^{1,\alpha}(B)$, the coefficients $\tilde{a}_{ij} = \partial_j F^i$ is Hölder-continuous. Thus we can use the perturbation theory to prove the $C^{1,\alpha}$ -continuous for the gradient of the solution.

Proposition 1.13. *In the assumption of Theorem 1.1, $u \in C^{2,\gamma}(B)$ for some $\gamma \in (0, 1)$.*

Proposition 1.14. $\tilde{a}_{ij} = \partial_j F_i \in C^\alpha(B)$.

Proof. For all $0 < r < 1$, by Theorem 1.11, there exists a constant $C > 0$ such that for all $x, y \in B_r$,

$$|\nabla u(x) - \nabla u(y)| \leq C |x - y|^\alpha$$

holds for some constant $\alpha \in (0, 1)$ independent of r .

For all $1 \leq i, j \leq n$, the function

$$F_{ij}(p) = \partial_j F_i(p) = \frac{\delta_{ij} \sqrt{1 + |p|^2} - p_i p_j / \sqrt{1 + |p|^2}}{1 + |p|^2}$$

has a property that $|\nabla F_{ij}|$ is bounded by $M > 0$ in $\{p \in \mathbb{R}^n : |p| \leq K\}$. Therefore, for all $x, y \in B_r$,

$$|F_{ij}(\nabla u(x)) - F_{ij}(\nabla u(y))| \leq |\nabla F_{ij}(\xi)| |\nabla u(x) - \nabla u(y)| \leq CM |x - y|^\alpha.$$

□

Now for any $1 \leq k \leq n$, $v = \partial_i u$ satisfies the following linear equation

$$\int_B \tilde{a}_{ij} \partial_j v \partial_i \phi = \int_B f \phi, \quad \forall \phi \in W_0^{1,2}(B).$$

To prove Proposition 1.13, we only need to prove $v \in C^{1,\alpha}(B)$.

In the proof below, as we often need to decompose the solution, we shall establish an estimate for weak solution of homogeneous equations with constant coefficients.

Lemma 1.15. *If $\{a_{ij}\}$ is a constant matrix satisfying the uniform elliptic condition, and suppose $u \in W^{1,2}(B)$ is a weak solution in the following sense:*

$$\int_B a_{ij} \partial_j u \partial_i \phi = 0, \quad \forall \phi \in W_0^{1,2}(B),$$

then there exists a constant $C = C(\Lambda, \lambda)$ such that for all $0 < \rho \leq R \leq 1$,

$$\begin{aligned} \int_{B_\rho} |u|^2 &\leq C \left(\frac{\rho}{R}\right)^n \int_{B_R} |u|^2, \\ \int_{B_\rho} |u - u_\rho|^2 &\leq C \left(\frac{\rho}{R}\right)^{n+2} \int_{B_R} |u - u_R|^2. \end{aligned}$$

Especially, since $\partial_k u$ is also a weak solution, we have

$$\int_{B_\rho} |\nabla u|^2 \leq C \left(\frac{\rho}{R}\right)^n \int_{B_R} |\nabla u|^2,$$

$$\int_{B_\rho} |u - (\nabla u)_\rho|^2 \leq C \left(\frac{\rho}{R} \right)^{n+2} \int_{B_R} |u - (\nabla u)_R|^2.$$

Here, $u_r = \frac{1}{|B_R|} \int_{B_R} u$.

First, we shall prove the following theorem:

Theorem 1.16. *For any $a_{ij} \in C^\alpha(B)$, $0 < \alpha < 1$ satisfying uniform elliptic condition, $f \in C^\infty(\bar{B})$, if $u \in W^{1,2}(B)$ is a weak solution in the following sense:*

$$\int_B a_{ij} \partial_j u \partial_i \phi = \int_B f \phi,$$

then $u \in C^\beta(B)$ for arbitrary $0 < \beta < 1$.

Proof. For any $0 < r < 1$, let $R_0 = \frac{1-r}{2}$, then for any $x_0 \in B_r$, $R \in (0, R_0)$, let $w \in W^{1,2}(B_R)$ such that $v = u - w \in W_0^{1,2}(B_R(x_0))$,

$$\int_{B_R(x_0)} a_{ij}(x_0) \partial_j w \partial_i \phi = 0, \quad \forall \phi \in W_0^{1,2}(B_R(x_0)),$$

then it is easy to check that for all $\phi \in W_0^{1,2}(B_R)$,

$$\int_{B_R(x_0)} a_{ij}(x_0) \partial_j v \partial_i \phi = \int_{B_R(x_0)} f \phi + \int_{B_R(x_0)} (a_{ij} - a_{ij}(x_0)) \partial_j u \partial_i \phi.$$

Then take test function $\phi = v$, we obtain

$$\begin{aligned} \int_{B_R(x_0)} |\nabla v|^2 &\leq \frac{1}{\lambda} \left(\int_{B_R(x_0)} |fv| + n^2 \|a\|_{C^\alpha(B_{\frac{1+r}{2}})} R^\alpha \int_{B_R(x_0)} |\nabla u| |\nabla v| \right) \\ &\leq \frac{1}{\lambda} \left(\|f\|_{L^{\frac{2n}{n+2}}(B_R(x_0))} \|v\|_{L^{\frac{2n}{n-2}}(B_R(x_0))} \right. \\ &\quad \left. + \frac{n^2}{2} \|a\|_{C^\alpha(B_{\frac{1+r}{2}})} R^\alpha \int_{B_R(x_0)} (|\nabla u|^2 + |\nabla v|^2) \right) \\ &\leq \frac{1}{\lambda} \cdot \|f\|_{L^\infty(B_R(x_0))} \omega_n^{\frac{n+2}{2n}} R^{\frac{n+2}{2}} \cdot C_{Sob} \|\nabla v\|_{L^2(B_R(x_0))} \\ &\quad + \frac{n^2}{2\lambda} \|a\|_{C^\alpha(B_{\frac{1+r}{2}})} R^\alpha \int_{B_R(x_0)} (|\nabla u|^2 + |\nabla v|^2) \\ &\leq \left(\frac{1}{4} + \frac{n^2}{2} \|a\|_{C^\alpha(B_{\frac{1+r}{2}}(x_0))} R^\alpha \right) \|\nabla v\|_{L^2(B_R(x_0))}^2 \\ &\quad + \frac{C_{Sob}^2}{\lambda^2} \|f\|_{L^\infty(B_R(x_0))}^2 \omega_n^{\frac{n+2}{n}} R^{n+2} \\ &\quad + \frac{n^2}{2\lambda} \|a\|_{C^\alpha(B_{\frac{1+r}{2}}(x_0))} R^\alpha \int_{B_R(x_0)} |\nabla u|^2. \end{aligned}$$

Therefore, there exists a constant $R_1 \in (0, R_0)$ such that if $R < R_1$, then

$$\int_{B_R(x_0)} |\nabla v|^2 \leq CR^{n+2} + CR^\alpha \int_{B_R(x_0)} |\nabla u|^2.$$

Then for any $0 < \rho < R < R_1$, we have

$$\begin{aligned} \int_{B_\rho(x_0)} |\nabla u|^2 &\leq 2 \int_{B_\rho(x_0)} |\nabla w|^2 + 2 \int_{B_\rho(x_0)} |\nabla v|^2 \\ &\leq 2C \left(\frac{\rho}{R}\right)^n \int_{B_R(x_0)} |\nabla w|^2 + 2 \int_{B_R(x_0)} |\nabla v|^2 \\ &\leq 4C \left(\frac{\rho}{R}\right)^n \int_{B_R(x_0)} |\nabla u|^2 + 6 \int_{B_R(x_0)} |\nabla v|^2 \\ &\leq \left(4C \left(\frac{\rho}{R}\right)^n + 6CR^\alpha\right) \int_{B_R(x_0)} |\nabla u|^2 + 6CR^{n+2}, \end{aligned}$$

so there exists a constant $R_2 \in (0, R_1)$ such that when $R < R_2$,

$$\int_{B_\rho(x_0)} |\nabla u|^2 \leq C \left(\frac{\rho}{R}\right)^n \int_{B_R(x_0)} |\nabla u|^2 + CR^{n+2}.$$

Thus by the iteration technique in Section 1.2.2, we obtain

$$\int_{B_R(x_0)} |\nabla u|^2 \leq CR^{n-2+2\alpha}, \quad \forall \alpha \in (0, 1).$$

Therefore, $u \in C^\alpha(B_{R_2}(x_0))$. And we can obtain $u \in C^\alpha(B)$. $\square$

For all $0 < r < 1$, $x \in B_r$, let $R_0 = \frac{1-r}{2}$, then for all $R \in (0, R_0)$, $B_R(x) \subset B_{\frac{1+r}{2}}$. We decompose $v = z + s$, where $s \in W^{1,2}(B_R(x))$ satisfying $s - v \in W_0^{1,2}(B_R(x))$, and

$$\int_{B_R(x)} \tilde{a}_{ij}(x) \partial_j s \partial_i \phi = 0, \quad \forall \phi \in W_0^{1,2}(B_R(x)),$$

and $z \in W_0^{1,2}(B_R(x))$.

Proof of Proposition 1.13. For all $0 < r < 1$, $x \in B_r$, let $R_0 = \frac{1-r}{2}$, then for all $R \in (0, R_0)$, $B_R(x) \subset B_{\frac{1+r}{2}}$. We decompose $v = z + s$, where $s \in W^{1,2}(B_R(x))$ satisfying $s - v \in W_0^{1,2}(B_R(x))$, and

$$\int_{B_R(x)} \tilde{a}_{ij}(x) \partial_j s \partial_i \phi = 0, \quad \forall \phi \in W_0^{1,2}(B_R(x)),$$

and $z \in W_0^{1,2}(B_R(x))$.

First we estimate the term ∇z . Using the technique in Section 1.2.2, we get $v, z \in C^\alpha(B_R(x))$. So for any $\rho \in (0, R)$, there exist a constant $\tilde{C}$ such that

$$\int_{B_\rho(x)} |\nabla v|^2 \leq C\rho^{n-2+2\alpha}, \quad \int_{B_\rho(x)} |\nabla z|^2 \leq C\rho^{n-2+2\alpha}.$$

Now for all $\phi \in W_0^{1,2}(B_R(x))$,

$$\int_{B_R(x)} \tilde{a}_{ij}(x) \partial_j z \partial_i \phi = \int_{B_R(x)} (\tilde{a}_{ij}(x) - \tilde{a}_{ij}(y)) \partial_j z(y) \partial_i \phi(y) dy + \int_{B_R(x)} f \phi.$$

For any $\rho \in (0, R)$, take test function $\phi = z\chi_{B_\rho(x)}$,

$$\begin{aligned} \int_{B_\rho(x)} |\nabla z|^2 &\leq \frac{1}{\lambda} \left(C\rho^\alpha \int_{B_\rho(x)} |\nabla v| |\nabla z| + \|f\|_{L^\infty} \int_{B_\rho(x)} |z| \right) \\ &\leq \frac{1}{\lambda} \left(\frac{C}{2} \rho^\alpha \int_{B_\rho(x)} |\nabla v|^2 + \frac{C}{2} \rho^\alpha \int_{B_\rho(x)} |\nabla z|^2 + \|f\|_{L^\infty} \int_{B_\rho(x)} |z| \right) \\ &\leq C' \left(\rho^\alpha \int_{B_\rho(x)} |\nabla z|^2 + \rho^\alpha \int_{B_\rho(x)} |\nabla z|^2 + \rho^{n+\alpha} \right) \\ &\leq C'' \left(\rho^{n-2+3\alpha} + \rho^\alpha \int_{B_\rho(x)} |\nabla z|^2 + \rho^{n+\alpha} \right) \\ &\leq C''' \left(\rho^{n-2+3\alpha} + \rho^\alpha \int_{B_\rho(x)} |\nabla z|^2 \right), \end{aligned}$$

where we use the Poincare's inequality.

Thus we obtain there exists a constant $R_1 > 0$ such that for all $\rho \in (0, R_1)$,

$$\int_{B_\rho(x)} |\nabla z|^2 \leq C\rho^{n-2+3\alpha}.$$

Now we investigate the property of ∇s . Notice that w is a weak solution of an elliptic equation with constant coefficients. We shall prove the following inequality:

$$\int_{B_\rho(x)} |\nabla s|^2 \leq C \left(\frac{\rho}{R} \right)^{n-2} \int_{B_R(x)} |\nabla s|^2, \quad \forall 0 < \rho < R < R_0$$

for some constant $C > 0$ independent of ρ, R, x .

Therefore, for all $0 < \rho < R < R_1$, we have

$$\begin{aligned}
& \int_{B_\rho(x)} |\nabla v - (\nabla v)_{x,\rho}|^2 \\
& \leq \int_{B_\rho(x)} |\nabla s - (\nabla s)_{x,\rho}|^2 + \int_{B_\rho(x)} |\nabla z|^2 \\
& \leq C \left(\frac{\rho}{R}\right)^{n-2} \int_{B_R(x)} |\nabla s - (\nabla s)_{x,R}|^2 + C \rho^{n-2+3\alpha} \\
& \leq C \left(\frac{\rho}{R}\right)^{n-2} \int_{B_R(x)} |\nabla v - (\nabla v)_{x,R}|^2 + C \left(\frac{\rho}{R}\right)^{n-2} \int_{B_R(x)} |\nabla z|^2 + C \rho^{n-2+3\alpha} \\
& \leq C \left(\frac{\rho}{R}\right)^{n-2} \int_{B_R(x)} |\nabla v - (\nabla v)_{x,R}|^2 + C \rho^{n-2} R^{3\alpha} + C \rho^{n-2+3\alpha}
\end{aligned}$$

Using the same iteration technique as in Section 1.2.2, for all $\beta \in (0, 3\alpha)$, we can finally obtain that there exists a constant $C_1 > 0$ (dependent of β) such that for all $R \in (0, R_1)$,

$$\int_{B_R(x)} |\nabla v - (\nabla v)_{x,R}|^2 \leq C_1 R^{n-2+\beta}.$$

Since we can take any $\alpha \in (0, 1)$, if $\alpha > \frac{2}{3}$, then

$$\int_{B_R(x)} |\nabla v - (\nabla v)_{x,R}|^2 \leq C_1 R^{n+2\gamma}$$

for some $\gamma \in (0, 1)$. Therefore, $v \in C^{1,\gamma}(B_{R_1}(x)) \implies v \in C^{1,\gamma}(B_r) \implies v \in C^{1,\gamma}(B)$. □

1.2.4 Step 4: From $C^{2,\alpha}$ to C^∞

Now since $u \in C^{2,\alpha}(B)$ for some $\alpha \in (0, 1)$, it becomes a classic solution, so we can use the Schauder's estimate to raise its regularity to smooth function.

Proposition 1.17. *In the assumption of Theorem 1.1, $u \in C^\infty(B)$.*

First, integrating by parts, we obtain

$$\sum_{i=1}^n \partial_i \left(\frac{\partial_i u}{\sqrt{1 + |\nabla u|^2}} \right) = f.$$

We rewrite it as

$$\tilde{a}_{ij} \partial_{ij} u = f, \quad \tilde{a}_{ij} = \frac{\delta_{ij}}{\sqrt{1 + |\nabla u|^2}} - \frac{\partial_i u \partial_j u}{\left(1 + |\nabla u|^2\right)^{\frac{3}{2}}}.$$

Now since we have $u \in C^{2,\alpha}(B)$, it is not difficult to prove that $\tilde{a}_{ij} \in C^{1,\alpha}(B)$. Moreover, if $u \in C^{k,\alpha}$ for any positive integer $k \geq 2$, then $\tilde{a}_{ij} \in C^{k-1,\alpha}(B)$. In fact, let $G_{ij} : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$G_{ij}(p) = \frac{\delta_{ij}}{\sqrt{1+|p|^2}} - \frac{p_i p_j}{(1+|p|^2)^{\frac{3}{2}}},$$

then for any multiple index λ , $|\lambda| = k-1$ for some positive integer $k \geq 2$,

$$\partial^\lambda G_{ij}(p) = \sum_{s=1}^n \partial_{p_s} G_{ij} \cdot \partial^\lambda \partial_s u,$$

where $\partial_{p_s} G_{ij}$ is bounded, and $\partial^\lambda \partial_s u \in C^k(B)$, so we obtain $\partial^\lambda G_{ij} \in C^{k-1,\alpha}(B)$.

Therefore, to prove the smoothness of the solution, we only need to prove the following proposition:

Proposition 1.18. *If we have $u \in C^{k+1,\alpha}(B)$ so $\tilde{a}_{ij} \in C^{k,\alpha}(B)$ for some positive integer $k \geq 1$ and $\alpha \in (0, 1)$, then $u \in C^{k+2,\alpha}(B)$.*

Proof. For all $0 < r < 1$, let $R_0 = \frac{1-r}{2}$, $r' = \frac{1+r}{2}$, then for all multiple index λ , $|\lambda| = k$, then by differentiation formally we have

$$\tilde{a}_{ij} \partial_{ij} \partial^\lambda u + \text{lower terms} = \partial^\lambda f.$$

Of course we cannot differentiate directly since we do not know whether the $(k+2)$ -th derivatives of u exist. So we shall first use the difference quotient method to obtain its existence.

We only need to prove for any multiple index τ , $|\tau| = k-1$ and any $1 \leq l \leq n$, $\partial_l \partial^\tau u$ exists and is continuous. For any function v , let

$$\Delta^h v(x) = \frac{v(x + h e_i) - v(x)}{h}.$$

Then

$$\begin{aligned} \partial^\tau (\tilde{a}_{ij} \partial_{ij} u) &= \partial^\tau f \\ \iff \tilde{a}_{ij} \partial_{ij} \partial^\tau u &= \partial^\tau f - \text{lower terms} \\ \iff \tilde{a}_{ij} \partial_{ij} v &= g, \end{aligned}$$

where $v = \partial^\tau u \in C^{2,\alpha}(B)$ and $g \in C^{1,\alpha}(B)$ is the difference of $\partial^\tau f$ and the lower terms in LHS. Using difference quotient, we get

$$\tilde{a}_{ij} (\Delta^h v) = \Delta^h g - (\Delta^h \tilde{a}_{ij}) \partial_{ij} \bar{v}, \quad \bar{v} = v(x + h e_i).$$

Using $g \in C^{1,\alpha}(B)$,

$$\Delta^h g(x) = \frac{1}{h} \int_0^1 \frac{d}{dt} g(x + t h e_i) dt = \int_0^1 \partial_i g(x + t h e_i) dt,$$

and we get $\Delta^h g \in C^\alpha(B_{r'})$ when $|h| < R_0$, and they have a uniform upper bound $\|\Delta^h g\|_{C^\alpha(B_{r'})} < C$. Similarly, since

$$\|\Delta^h \tilde{a}_{ij}\|_{C^\alpha(B_{r'})} \leq C,$$

$$\|\bar{v}\|_{C^\alpha(B_{r'})} \leq C$$

when $|h| < R_0$, by $C^{2,\alpha}$ - a priori estimate, we have

$$\|\Delta^h v\|_{C^{2,\alpha}(B_r)} \leq C\|\Delta^h g - (\Delta^h \tilde{a}_{ij})\partial_{ij}\bar{v}\|_{C^\alpha(B_{r'})} + C\|\Delta^h v\|_{C^\alpha(B_{r'})} \leq C,$$

therefore, $\{\Delta^h v\}$ is a uniformly bounded and equi-continuous family of functions in $C(\bar{B}_r)$, and by Arzela-Ascoli's theorem, $\Delta^h v \rightarrow Q$ uniformly for a unique function $Q \in C(\bar{B}_r)$. Therefore, $\partial_t \partial^\tau u$ exists and equals to Q . Eventually, we get $u \in C^{k+2}(B_r)$ for any $0 < r < 1$. Now for any multiple index λ , $|\lambda| = k$, We have

$$\begin{aligned} \partial^\lambda (\tilde{a}_{ij} \partial_{ij}) &= \partial^\lambda f \\ \iff \tilde{a}_{ij} \partial_{ij} U &= F, \quad U = \partial^\lambda u, \quad F = \partial^\lambda f - \text{lower terms} \end{aligned}$$

and note that RHS is in $C^\alpha(B)$. We use the existence and regularity of Dirichlet problems, so we conclude that $U \in C^{2,\alpha}(B_r)$, which gives $u \in C^{k+2,\alpha}(B_r)$ for any $0 < r < 1$. Thus, we have $u \in C^{k+2,\alpha}(B)$. $\square$

Proposition 1.17 can be directly proved by Proposition 1.18.

What we need is the Schauder estimate and the existence and unique theorem for linear elliptic PDE:

Theorem 1.19 (A priori estimate, Gilbarg & Trudinger, Corollary 6.3). *For all bounded domain $\Omega' \subset \mathbb{R}^n$, and $\Omega \Subset \Omega'$, if $u \in C^2(\Omega')$ satisfies*

$$a_{ij} \partial_{ij} u = f \text{ in } \Omega',$$

where $a_{ij} \in C^\alpha(\Omega')$ satisfies the uniformly elliptic condition in Ω' , $f \in C^\alpha(\Omega')$ and their C^α -norm in Ω' are finite, then there exists a constant $C = C(\lambda, \Lambda, \Omega, \Omega')$ such that

$$\|u\|_{C^{2,\alpha}(\Omega)} \leq C (\|f\|_{C^\alpha(\Omega')} + \|u\|_{C^\alpha(\Omega')}).$$

Theorem 1.20 (Gilbarg & Trudinger, Theorem 6.19). *For all bounded domain $\Omega \subset \mathbb{R}^n$ whose boundary is C^∞ , assume $a_{ij} \in C^\alpha(\bar{\Omega})$, $f \in C^\alpha(\bar{\Omega})$, $\phi \in C^0(\bar{\Omega})$, then the Dirichlet problem:*

$$a_{ij} \partial_{ij} u = f \text{ in } \Omega, \quad u = \phi \text{ on } \partial\Omega$$

has a unique solution $u \in C^{2,\alpha}(\Omega) \cap C^0(\bar{\Omega})$.

Moreover, if $a_{ij}, f \in C^{k,\alpha}(\bar{\Omega})$ for any positive integer k , then $u \in C^{k+2,\alpha}(\bar{\Omega})$.

1.3 Boudary Regularity

Now we are going to prove the global result - Theorem 1.2. Now we already have $u \in C^\infty(\Omega)$. If we can prove that $u \in C^{1,\alpha}(\bar{\Omega})$, then $\tilde{a}_{ij} \in C^\alpha(\bar{\Omega})$, so by Theorem 1.20, $u \in C^{2,\alpha}(\bar{\Omega})$. Repeat this operation, we get $u \in C^\infty(\bar{\Omega})$.

Let $\mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_n > 0\}$, $B^+ = B_1 \cap \mathbb{R}_+^n$.

Theorem 1.21 (Gilbarg & Trudinger, Theorem 9.31). *Suppose that $u \in W_{loc}^{2,n}(B^+) \cap C^0(\bar{B}^+)$ satisfies the equation*

$$a_{ij}\partial_{ij}u = f,$$

where $f \in L^\infty(B^+)$, $u = 0$ on $\bar{B}^+ \cap \partial\mathbb{R}_+^n$, then for any $R \leq 1$, we have

$$\text{osc}_{B^+} \frac{u}{x_n} \leq CR^\alpha \left(\text{osc}_{B^+} \frac{u}{x_n} + \sup_{B^+} |f| \right),$$

or

$$\text{osc}_{B^+} \frac{u}{x_n} \leq CR^\alpha \left(\sup_{B^+} |\nabla u| + \sup_{B^+} |f| \right),$$

where $C > 0$, $\alpha \in (0, 1)$ only depend on n, λ, Λ .

Thus, we obtain $u \in C^{1,\alpha}(\bar{B}^+)$ for some $\alpha \in (0, 1)$.

Proof of Theorem 1.2. By $u \in W^{1,\infty}(\Omega)$, we have $u = \phi$ in $\partial\Omega$. First we shall prove the theorem in the case of $\phi = 0$. For any $x_0 \in \partial\Omega$, there always exists a C^∞ straightening map φ that maps a certain neighborhood U of x_0 (within Ω) into the upper hemisphere B^+ , and maps $\bar{U} \cap \partial\Omega$ into $\bar{B}^+ \cap \partial\mathbb{R}_+^n$. Consider $\tilde{u} = u \circ \varphi^{-1}$ satisfies

$$\tilde{a}_{sl}^* \partial_{sl} \tilde{u} = \tilde{f}, \quad \tilde{a}_{sl} = \tilde{a}_{ij}^* \frac{\partial \varphi^s}{\partial x_i} \frac{\partial \varphi^l}{\partial x_j}, \quad \tilde{f} = f \circ \varphi^{-1} - \tilde{a}_{ij} \frac{\partial^2 \varphi^l}{\partial x_i \partial x_j} \partial_l \tilde{u}.$$

Obviously $\left| \tilde{f} \right|_{L^\infty(\bar{B}^+)}$ is bounded, because $\partial_l \tilde{u}$ is bounded by a constant multiple of ∇u , which is bounded. We can easily check that $\tilde{a}_{sl}^*$ satisfies the uniform elliptic condition, and by Theorem 1.21, $\tilde{u} \in C^{1,\alpha}(\bar{B}^+)$ for some $\alpha \in (0, 1)$, and $u \in C^{1,\alpha}(\bar{U})$. Covering $\partial\Omega$ by finite closures of neighborhood, we finally get $u \in C^{1,\alpha}(\bar{\Omega})$ for some $\alpha \in (0, 1)$. Since $u \in C^{1,\alpha}(\bar{\Omega})$, we get $\tilde{a}_{ij} \in C^\alpha(\bar{\Omega})$, then by Theorem 1.20, $u \in C^{2,\alpha}(\bar{\Omega})$. Then $\tilde{a}_{ij} \in C^{1,\alpha}(\bar{\Omega})$, and repeat the operation, we get $u \in C^{3,\alpha}(\bar{\Omega})$. Finally, we obtain $u \in C^\infty(\bar{\Omega})$. $\square$